Homework 3 sols:

B. Sylow p-group 之相关: Sylow's 3 theorem.

1. $|G:N_G(S)| \equiv 1 \pmod p$ Step 1: $N_G(S)$ 至少2个 Sylow p-子群

$S \curvearrowright G/N_G(S)$. for $\forall gN_G(S) \neq N_G(S)$. ($g \notin N_G(S)$ s.t.)

$\mathrm{Stab}_G(gN_G(S)) \neq S$. otherwise. $sgN_G(S) = gN_G(S)$ $\forall s \in S$

$\Rightarrow g^{-1}sg \in N_G(S)$ $\forall s \in S$ then $g^{-1}Sg \subseteq N_G(S)$

$g \notin N_G(S) \Rightarrow g^{-1}Sg \neq S \Rightarrow N_G(S)$ 有至少2个 Sylow p-group

But in $N_G(S)$. $\forall g \in N_G(S)$ $g^{-1}Sg = S$ contradict with $\forall$ sylow p-group conjugates

- since $|G:N_G(S)| \equiv n_p \equiv 1 \pmod p$

↑ number of p-sylow group of $N_G(S)$ Rmk: Can be generalized to $\forall$

U s.t. $N_G(S) \leq U \leq G \Rightarrow |G:U| \equiv 1 \pmod p$

Follow from $|G:N_G(S)| = |G:H|\,|H:N_G(P)|$

$= |G:H|\,|H:N_H(P)| \Rightarrow$ modulo p

$N_G(P) \cap H = N_G(P)$ since $N_G(P) \leq H$

2. $|G| < +\infty$. $S < G$ Sylow p-grp

$H \lhd G$. $S \cap H \subseteq H$ sylow p-group

using prop24: $\exists g \in G$ s.t. $gSg^{-1} \cap H \supseteq H$ Sylow p-grp

and. $S \cap H = g^{-1}(gSg^{-1} \cap H)g = S \cap gHg^{-1}$

so we know $gSg^{-1} \cap H$ conjugate to $S \cap H$

then • $S \cap H$ is sylow p-grp of H

- $|G| < +\infty$. $H \lhd G$ $\pi: G \to G/H$
  $S < G$ Sylow p-group. $\Rightarrow S \cap H$ sylow p-group of H then order of sylow p-group of G/H is. $\frac{|S|}{|S \cap H|}$

- Aim: Show $|\pi(S)| = \frac{|S|}{|S \cap H|}$ $\pi: G \to G/H$
  $s \mapsto \pi(s)$. then. $\forall s \in S$ $s \in \ker\pi \Leftrightarrow s \in S \cap H$

$\Rightarrow \pi(S)$ is group $|\pi(S)| = \frac{|S|}{|S \cap H|} \Rightarrow$

$\pi(S)$ is sylow p-grp of G/H

$S' < G/H$ sylow $\Rightarrow$. $|\pi^{-1}(S')| = |S'| \cdot |H|$ $\rightsquigarrow$ $|\pi^{-1}(S')|$ has p-order ($\mathrm{ord}_p|\pi^{-1}(S')| = \mathrm{ord}_p|G|$)

$\Rightarrow$ consider sylow p-grp of $\pi^{-1}(S')$ denoted by S.

Aim: $\pi(S) = S'$ $\quad \pi|_S : S \longrightarrow S'$

$s \in S, \longmapsto sH.$

$$|\mathrm{Im}\pi|_S| = \frac{|S|}{|\ker \pi|_S|} = \frac{|S|}{|H \cap S|} = |S'| \Rightarrow \mathrm{Im}\pi|_S = S' \quad \square$$

Shows

$\{$sylow p-group of $G\}$ $\longleftrightarrow$ $\{$sylow p-group of $G/H\}$

S $\longmapsto$ $\pi(S)$

$S^* :=$ sylow p-group of $\pi^{-1}(S')$ $\longleftarrow$ S'

B4) Frattni-Technique:

$G \curvearrowright X$. $H < G$ subgrp.

$\Downarrow$ induce

$H \curvearrowright X$. $\rightsquigarrow$ suppose trans.

$\forall x \in X$ $g \in G$.

$\exists x' \in X$ s.t. $gx = x'$

$\exists h \in H$ s.t. $hx = x' \Rightarrow gx = hx. \Rightarrow$

$h^{-1}gx = x \Rightarrow$.

$h^{-1}g \in \mathrm{Stab}_G(x). \Rightarrow g \in h \cdot \mathrm{Stab}_G(x)$

$\Rightarrow g \in H\,\mathrm{Stab}_G(x)$

$\Rightarrow G = H \cdot \mathrm{Stab}_G(x)$.

b). Technique:

$X = \{$Sylow p-grp of $H\}$ $S \in X$ element in X ($S < H$ is sylow p-grp of H)

$H \curvearrowright X$ trans. $\Rightarrow$ (using A). $G = H \cdot \mathrm{Stab}_G(S)$.

$\Uparrow$ need to show $G \curvearrowright X$ well defined.

(key: using $X = \{$sylow p-group of $H\}$)

$S \in X$

$\mathrm{Stab}_G(S) \overset{\text{def}}{=} N_G(S)$. $\Rightarrow |gSg^{-1}| = |S| \,\big|\, |H|$

$g \in G$ $S \in X \Rightarrow g \cdot S := gSg^{-1} \in gHg^{-1}$ since H normal

$= H$

then by $S \in X \Rightarrow gSg^{-1} \in X \Rightarrow G \curvearrowright X$ well-defined

B5). $|G| < \infty$. $S < G$ sylow p-grp.

$N_G(S) \leq H < G \Rightarrow$. $N_G(H) = H$.

$S \leq N_G(S) \leq H \leq N_G(H) < G$

$\Rightarrow S$ since S is sylow p-

$X := \{$sylow p-grp of $H\}$ $H \trianglelefteq N_G(H)$. gr

- $N_G(H) \curvearrowright X \Leftarrow H \curvearrowright X$ transitive

transitive.

$\Downarrow$

$N_G(H) = H$. $N_G(S) = H$ then $N_G(H) = H$

B6): fix $H = N_G(S)$. $\checkmark$

Be careful: Frattni-Technique:

A). order 60 simple group:

A1) $60 = 2^2 \times 3 \times 5$.

$S_2 := \#\{\text{sylow 2-group}\} \Rightarrow S_2 \mid 3\times 5 \;\; S_2 \equiv 1 \pmod 2 \Rightarrow S_2 \in \{1,3,5,15\}$

$S_3 := \#\{\text{sylow 3-grp}\} \Rightarrow S_3 \mid 4\times 5 \;\; S_3 \equiv 1 \pmod 3 \Rightarrow S_3 \in \{1,4,10\}$.

$S_5 := \#\{\text{sylow 5-group}\} \Rightarrow S_5 \mid 4\times 3 \;\; S_5 \equiv 1 \pmod 5 \Rightarrow S_5 \in \{1,6\} \;\; S_5 \neq 1 \Rightarrow S_5 = 6$

A2) $5 \mid |H| \Rightarrow \exists$ 5-sylow grp P_1 of G s.t. it is subgrp of H. and $H \lhd G \Rightarrow gP_1g^{-1} \subseteq H$.

and $G \curvearrowright \{\text{sylow 5-grp of } G\}$ trans

$\Rightarrow H$ contains all sylow 5-group.

and $H_i \cap H_j = 1 \;\forall i \neq j$ since $|H_i| = |H_j| = 5 \Rightarrow |H| \geq (4)\times 6 + 1 = 25$

$\Rightarrow |H| = 30$ since $|H| \mid |G|$ and $|H| < |G|$

$|H| = 30$ Claim: $|H|$ has only 1 5-subgrp $\Longleftrightarrow$ H contains normal sylow 5-subgrp.

suppose not.; then $\exists$ normal 2-subgrp / normal 3-subgrp.

otherwise $S_2 \geq 3 + S_3 \geq 4 + S_5 = 6$

$\Downarrow$

$|H| \geq 1 + (3\times 1) + 1 + (4\times 2) + 1 + 6\times 4 = 4 + 9 + 15 > 30$. contradiction.

$\Rightarrow$ H has normal subgrp M ($|M| = 2$ or 3)

$\Rightarrow H/M$ has order $15/20$, and H/M has 1 normal 5-subgrp (otherwise must ≥ 6 sylow group ×)

suppose $M_2 \lhd H/M$ a 5-sylow normal subgrp.

$\Downarrow$ by correspondence of quotient group.

H has only 1 5-sylow normal subgrp

A3) $|G/H| = 30, 20, 15 \overset{A2}{\Rightarrow}$ has only 1 sylow 5-subgrp.

$G \xrightarrow{\pi} G/H$. sylow 5-subgrp $\cdot N' \Rightarrow |\pi^{-1}(N')| \in \{10, 15, 20\} \leq |G|$ let $H = \pi^{-1}(N')$

$\pi^{-1}(N') \quad N'$ ∎

A4) • $|H| = 6$. sylow: $S_3 \in \{1, 4\}$. $S_2 \in \{1, 3\}$

• $|H| = 12$ $S_3' \in \{1, 4\}$ $S_2' \in \{1, 3\} \Rightarrow S_3' = 1$ or $S_2' = 1 \Rightarrow G$ has order 2 or 3 normal subgrp $\Longrightarrow$ back to A3)

when $S_3' = 4$ $\Uparrow S_2' = 3$ has at least $4\times 2 + 1 + 3\times 1 + 1 = 9 + 4 = 13 > 12$ elements ↯ gives contradiction

—— G: $|G| = 60$ + Simple

A5) Suppose $H < G$ + G simple. $\Rightarrow S_5 = 6$. $S_2 \in \{3, 5, 15\}$ $S_3 \in \{4, 10\}$

$G \curvearrowright G/H \Rightarrow \varphi : G \longrightarrow S_{G/H}$ $\ker(\varphi)$ nontrivial ($\ker\varphi \neq 60$)

$\Downarrow$ ~ Since $|G/\ker\varphi|$ simple

$|G| = 60$ $|\ker\varphi| = 1$

$\Rightarrow G \hookrightarrow S_{G/H} \Rightarrow |G/H| \geq 5$ if "=" holds $\Rightarrow$ "✓"

$|G| = 60$

$$\frac{|S_{G/H}|}{|G|} = \frac{120}{60} = 2$$

then $G = A_5$ (since order 2 subgrp of S_5 must be A_5)

A6) $S_5 = 6$ ✓

$X_2 := \{2\text{-sylow grp of } G\}$

$X_3 := \{3\text{-sylow grp of } G\}$

$\Rightarrow G \curvearrowright X_2, X_3 \Rightarrow \varphi_2 : G \longrightarrow S_{X_2}$

$\varphi_3 : G \longrightarrow S_{X_3}$

$\Rightarrow$ Since $G \curvearrowright X_2 . X_3$ trans.

$\Rightarrow \mathrm{Im}\varphi_2 . \mathrm{Im}\varphi_3 \neq \{e\}$

G simple.

$\Rightarrow \{\ker\varphi_2 . \ker\varphi_3 = \{e\}\}$

$\Rightarrow |S_{X_2}| . |S_{X_3}| \geq 60.$

$\Rightarrow |X_2| . |X_3| \geq 5$

A7)

$S_2 = 5 \Rightarrow G \curvearrowright X_2$ (conjugate.) $\rightsquigarrow \varphi_2 : G \hookrightarrow S_{X_2}$

$$\Rightarrow \frac{|S_{X_2}|}{|G|} = 2 \Rightarrow \frac{|S_{X_2}|}{|\mathrm{Im}\varphi_2|} = 2$$

then. $\mathrm{Im}\varphi_2 \lhd S_{X_2}$ and then $\mathrm{Im}\varphi_2 \simeq A_5 \simeq G$

$\Rightarrow . S_2 \in \{5, 15\}$ $S_3 = 10$

A8) $S_2 = 15$ if # sylow 2 grp P, Q s.t. $|P \cap Q| = 2$

$\Rightarrow . \ |G| \geq (4-1) \times 15 + (3-1) \times 10 + (5-1) \times 6 > 60.$ ×

So. $\exists . P, Q . \ |P \cap Q| = 2.$ and $P \subseteq N_G(P \cap Q)$ (since P, Q com) $\Rightarrow . \ 4 \mid |N_G(P \cap Q)|$

and $|N_G(P \cap Q)| > 4 + \neq G$

$\subsetneq G$ normal

then • $[G : N_G(P \cap Q)] \geq 5 \Rightarrow |N_G(P \cap Q)| \leq 12$

$+ \ 4 \mid |N_G(P \cap Q)|$

$\Rightarrow N_G(P \cap Q) = 12$

$\Rightarrow [G : N_G(P \cap Q)] = 5$

A9)

$S_2 = 5$ ✓ $S_2 = 15 \xrightarrow{A8} \exists H < G.$

$H \neq G$. s.t. $[G : N_G(P \cap Q)] = 5 \xrightarrow{A5} G \simeq A_5$

pick $H = N_G(P \cap Q)$.

$S_2 = 5$ has the form $P = \{e, (12)(34), (13)(24), (14)(23)\}$. ($C_5^4$)